Degrees of Carbon Functional Series

A domain-level series on what the platform measures, how it analyses it, how the results compare with authoritative climate references, and what its trends imply for the Paris Agreement targets. It complements the Technical Series, which covers how the platform is built. No programming is needed to follow it.

> **Status: complete as of October 2026.** All five sections are written. Facts about pledges and the COP process are dated; see the "as of" note at the top of each document.

1. The domain in one page

The greenhouse effect. Gases such as carbon dioxide absorb heat that the Earth would otherwise radiate to space. Burning fossil fuels, producing cement, farming and changing how land is used add these gases to the atmosphere faster than natural processes remove them, so their concentrations rise and the surface warms.

The chain this platform follows. Each link has its own measurements, and the platform holds data for each:

```
emissions  ⟶  atmospheric concentration  ⟶  global temperature
(what we release)   (what stays in the air)        (the physical response)
```

Link	Question	Platform data
Emissions	How much is released, by whom, of which gas?	OWID / Global Carbon Project (CO_2), PRIMAP-hist (all main gases)
Concentration	How much has accumulated in the air?	NOAA Mauna Loa (measured) spliced to Law Dome ice cores
Temperature	How much has the surface warmed?	Berkeley Earth, relative to 1850–1900

The platform's forecasts and scenarios apply to the first link; its relationship analysis (Methodology) connects the three.

The gases. Carbon dioxide (CO_2) dominates by volume. Methane (CH_4), nitrous oxide (N_2O) and the fluorinated gases ("F-gases") are much less abundant but trap far more heat per tonne. To add them up, each is converted to **CO_2-equivalent (CO_2e)** with a *global warming potential* (GWP) over a fixed horizon. This platform uses the IPCC Fifth Assessment Report's 100-year values (CH_4 = 28, N_2O = 265) wherever it reports a combined total. The choice of GWP set changes the totals, so any comparison with another source has to check which one it used.

Who is counted. Emissions inventories are *territorial*: they count emissions released within a country's borders, not those embodied in the goods it imports. They usually exclude international aviation and shipping, and treat land use, land-use change and forestry (LULUCF) separately because it is far more uncertain than fossil emissions. Each document says which convention a number follows.

2. What this platform is, and is not

It is	It is not
A transparent, reproducible analysis of published historical data	A climate model; it does not simulate physics
Short-horizon statistical forecasts of CO_2 with held-out testing	A prediction of what governments or technology will do
Illustrative what-if scenarios (constant percentage reductions)	Calibrated policy pathways
A way to see trends and compare them with reference benchmarks	An authority; the IPCC and the sources in Benchmarks are

Keep this table in mind when reading the Findings section.

3. Key numbers

These are the platform's own latest values, as of the data refresh behind this document. Later sections refer back to them. Sources, scope and caveats are in the Data Sources documents (emissions, concentrations and temperature, quality).

Quantity	Latest value	Source and scope
World CO_2 from fossil fuels and cement, 2024	38,599 $MtCO_2$	OWID / Global Carbon Project; includes international aviation and shipping
World land-use-change CO_2, 2024	4,586 $MtCO_2$	Same source; a modelled estimate with large uncertainty
World total GHG, 2024	50,550 $MtCO_2e$	PRIMAP-hist; CO_2, CH_4, N_2O, F-gases (AR5 GWP-100); excludes LULUCF and international transport
Share of that total, 2024	CO_2 75.1%, CH_4 16.7%, N_2O 5.4%, F-gases 2.9%	PRIMAP-hist
Atmospheric CO_2, 2025 annual mean	427.35 ppm	NOAA Mauna Loa
Global temperature, 2025	1.45 °C above 1850–1900 (preliminary)	Berkeley Earth; 95% uncertainty ±0.04 °C is statistical only
Global temperature, 2024	1.55 °C above 1850–1900 (preliminary)	Berkeley Earth

Fossil CO_2 is not the same thing as total greenhouse gases, and the two sources above disagree by a few percent on shared quantities because of scope and method; the platform discloses that and does not reconcile it.

4. The series

#	Section	What it covers
1	Data Sources: emissions inventories, concentrations and temperature, quality and harmonization	The five sources, their coverage, quality and harmonization
2	Methodology: trend analysis, forecasting, emissions–temperature relationship, scenarios and temperature mapping, limitations and uncertainty	How the numbers are produced, and how far they can be trusted

#	Section	What it covers
3	Benchmarks: reference landscape, emissions, temperature and concentration, pathways, carbon budgets and warming outlooks	The platform's outputs against IPCC, UNEP, the Global Carbon Budget and other references
4	Paris Agreement: the agreement and COP process, targets as numbers, country NDCs	The agreement, its targets expressed as numbers, and country NDCs
5	Findings: where things stand, trajectory against Paris, country differences, caveats and reading guide	Where emissions stand, how the scenarios compare with Paris-aligned pathways, and caveats

Glossary

Term	Meaning
Anomaly	A deviation from a reference period's average, not an absolute temperature
CO_2e	CO_2-equivalent: other gases converted to CO_2 using a global warming potential
GWP	Global warming potential: heat trapped per tonne relative to CO_2, over a fixed horizon (here 100 years)
LULUCF	Land use, land-use change and forestry
Mt, Gt	Million tonnes, billion tonnes (1 Gt = 1,000 Mt)
ppm	Parts per million: a concentration, as in CO_2 molecules per million air molecules
Pre-industrial	Conventionally 1850–1900 in this platform's temperature data
Territorial emissions	Emissions released within a country's borders